

Fertility and Housing Tenure Choice: Model and Population Closure

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Experimental repaired-E5 (floor) specification — not the promoted benchmark

This Talk

One-market national lifecycle model of housing and fertility. Since May:

- **Fertility is sequential:** birth-by-birth attempts under declining fecundity, with children ever born and children at home tracked separately
- **Measured income risk:** after-tax persistent earnings process fixed outside the model
- **One housing market** (the two-location structure is set aside for the national question)

Today's focus:

- The **population closure**: what population-level statement can the model honestly make when fertility is endogenous and below replacement?
- Corrected **policy results** under the new closure

Model

Environment

- **Agents.** A mass S of households; age $j = 1, \dots, 17$ four-year periods from age 18; retirement at 65; survival risk in old age
- **Children.** Two state variables per household: n children ever born, m children currently at home; each child at home matures independently with per-period probability $2/9$ (18 expected years at home); two matured children form one new potential household
- **Endowments.** Entrants start as childless renters on a fixed entry-wealth distribution; persistent after-tax earnings risk (next slide); exogenous wage level and interest rate
- **Markets.** One housing market: rental services and owned houses (ladder); one-period bond; no aggregate risk
- **Institutions.** Property tax τ_H with a balanced-budget lump-sum transfer; payroll tax and stationary pension; warm-glow bequests — estates are valued by parents but transfer to no one
- **Entry.** New households enter at age 18: retained matured city-born children plus an outside inflow (the closure; Section 3)

Preferences

Flow utility over consumption and housing services, scaled by family size:

$$u(c, \mathbf{h}; m) = \frac{\left[(c/e_c(m))^\alpha (\mathbf{h}/e_h(m))^{1-\alpha} \right]^{1-\sigma}}{1-\sigma} + \psi_{\text{child}} m, \quad \sigma = 2,$$

with equivalence scales $e_c(m)$, $e_h(m)$ increasing in children at home.

- **Owner premium.** Housing services are χh for owners ($\chi > 1$) and h^R for renters
- **Child space floor** (this arm): each child at home requires $\underline{h}$ rooms — the estimated floor is 0.61 rooms per child
- **Bequests.** Warm glow at death over terminal wealth $b + ph$, curvature shifted by (θ_0, θ_1)

The estimated child flow utility ψ_{child} is zero at this winner: children are valued through the choice structure, not a flow term.

Sequential Fertility

Births arrive one at a time through an attempt margin under biological limits:

- In the fecund window, a household chooses each period whether to **attempt** a birth: a discrete choice with taste shocks (scale κ_f for the first birth, κ_f^{cont} for subsequent births)
- An attempt succeeds with an age-declining **fecundity** probability — timing is chosen, arrival is risky
- A birth moves the household $(n, m) \rightarrow (n + 1, m + 1)$; children at home raise the equivalence scales and the room floor, then mature out

The margin this delivers: households postpone attempts when housing is expensive and run into the fecundity clock — the mechanism connecting housing costs to completed fertility and childlessness.

Earnings and Institutions

After-tax earnings are fixed outside the model from measured objects:

$$\log y_{j+1} = \rho \log y_j + \varepsilon, \quad \rho = 0.914, \quad \sigma_{\varepsilon}^{\text{annual}} = 0.169,$$

a five-state Rouwenhorst discretization, aggregated to four-year periods.

- Persistence and innovation variance from the panel literature; progressivity retention $1 - \tau = 0.819$ (HSV, $\tau = 0.181$)
- Age profile of earnings; payroll tax 0.179 funds a stationary pension
- Property tax τ_H (baseline 1% annual) rebated lump-sum — the policy instrument later

Income risk is measurement, not calibration: no earnings parameter is estimated.

Housing, Tenure, and Budget Sets

A household either rents or owns:

$$h^R \in (0, \bar{h}^R], \quad h \in \{H_1, \dots, H_6\} = \{2, \dots, 11\} \text{ rooms}, \quad \bar{h}^R = 6.$$

Renters pay rent rh^R , save $b' \geq 0$. **Owners** post a down payment and borrow against the house:

$$b' \geq -\phi ph, \quad \phi = 0.80 \text{ (20\% down)},$$

pay maintenance δ , property tax τ_H , and a transaction cost ψ when adjusting.

- **Segmentation:** the largest rental is smaller than the largest owned unit, so family-sized space is reached through ownership — the down payment is the barrier a birth must clear

Housing Supply and Equilibrium

Supply is an upward-sloping curve in **levels**, and rents are tied to prices by user cost:

$$H^S = H_0 \left(\frac{r}{\bar{r}} \right)^\xi, \quad r = (q + \delta + \tau_H) p.$$

- One market-clearing condition: aggregate services demanded = H^S
- The level curve is the fixed-land statement: a permanently larger population permanently raises rents — this is what makes the population question bite
- Stationary equilibrium: prices, the transfer, and the household distribution constant; the government budget clears

Wages and the interest rate stay exogenous: the housing market is the only price feedback.

The Household's Problem

Within a period: attempt decision, then tenure, housing, and saving:

$$V_j(b, h, y, n, m) = \mathbb{E}_\varepsilon \max_{a \in \{0,1\}} \left\{ \underbrace{W_j^a(b, h, y, n, m)}_{\text{tenure/housing/saving problem}} + \varepsilon_a \right\},$$

where an attempt ($a = 1$) succeeds with fecundity π_j and, if successful, delivers $(n + 1, m + 1)$ next period. The inner problem chooses rent-vs-own, the housing quantity or rung, consumption, and b' subject to the budget and collateral constraints, with deterministic tenure choice.

Old age: no attempts, survival risk, warm-glow bequest at death; matured children leave the state vector and join the entrant pool.

Calibration

Externally Fixed Objects

	Object	Value	Source
<i>Pref.</i>	σ risk aversion	2.0	standard
<i>Fin.</i>	q interest (annual)	0.04	standard
	δ depreciation (annual)	0.02	Greaney et al.
	τ_H property tax (annual)	0.01	US average
	ϕ financed share	0.80	20% down payment
	ψ transaction cost	0.06	Greaney et al.
<i>Life</i>	17 four-year periods	ages 18–85	retirement 65
<i>Income</i>	$\rho, \sigma_\varepsilon, \text{HSV } \tau$	0.914, 0.169, 0.181	FL $\times$ HSV
	τ_{pay} payroll	0.179	Greaney et al.
<i>Children</i>	maturation hazard	2/9 per period	18 expected years
<i>Housing</i>	ladder, rental cap	$\{2, \dots, 11\}$, 6 rooms	room units

External restriction on the search: annual $\beta \geq 0.94$.

Estimated Parameters (floor arm)

Param	Description	Estimate	Bounds	Near bound?
β (annual)	Discount factor	0.9946	[0.94, 0.9995]	no
ψ_{child}	Child flow utility	0.000	[0, 3]	at lower
κ_f	Attempt taste scale, first birth	2.987	[0.02, 50]	no
κ_f^{cont}	Attempt taste scale, later births	1.713	[0.02, 50]	no
χ	Owner service premium	1.037	[0.1, 5]	no
H_0	Housing supply intercept	9.873	[0.2, 80]	no
θ_0	Bequest strength	0.159	[0, 8]	near lower
θ_1	Bequest shifter	0.094	[0.02, 16]	near lower
$\underline{h}$	Child room floor	0.614	[0.1, 1.8]	no

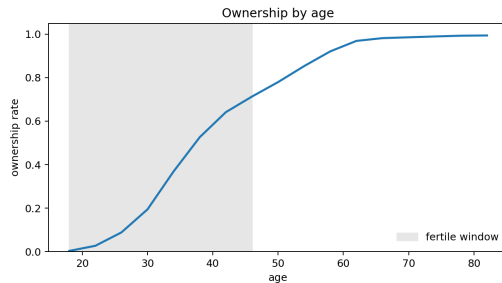
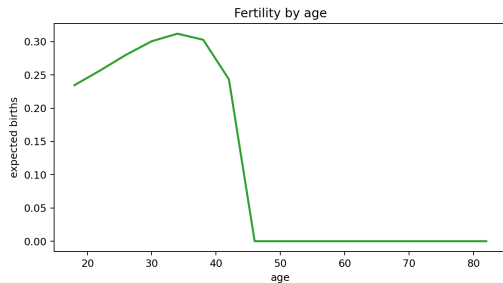
Nine free parameters against twelve moments. The bequest block sits near its lower bounds and the child flow utility is exactly zero: saving-for-bequests and direct child utility are doing little work at this winner.

Target Fit (floor arm; loss 297.1)

	Moment	Target	Model	Gap	Contr.
<i>Fertility</i>	Completed fertility	1.918	1.967	+0.049	3.4
	Childlessness	0.188	0.104	-0.084	121.5
	Mean age at first birth	25.31	25.57	+0.26	1.1
	First births at 30+	0.270	0.327	+0.057	51.5
<i>Housing</i>	Rooms response to first birth	0.664	0.690	+0.026	0.6
	Rooms, 3+ vs. 1-2 children	0.368	0.359	-0.009	0.2
	Family ownership gap	0.168	0.162	-0.006	0.5
	Ownership rate	0.575	0.582	+0.007	0.1
	Aggregate mean rooms	5.780	6.478	+0.698	5.8
<i>Wealth</i>	Wealth / earnings	6.873	5.794	-1.079	7.3
	Bequest flow / wealth	0.0088	0.0087	-0.0001	0.0
	Old wealth p90/p50	3.448	2.090	-1.358	105.1

Three rows carry 94% of the loss: childlessness, old-age wealth dispersion, late first births. An overnight 16-chain multi-start on the unchanged contract (fresh seeds; 24 chains total) finds nothing better: these are **mechanism limits, not search failures**.

Lifecycle Fit



Standard diagnostic packet; full set in the appendix and in the packet folder.

Population Closure

The Closure Problem

With endogenous below-replacement fertility, a closed stationary population cannot exist:

$$\underbrace{B_0 \approx 0.053}_{\text{matured city-born flow}} < \underbrace{E_0 \approx 0.062}_{\text{entrants needed for stationarity}} \Rightarrow \text{natives replace } \sim 86\% \text{ of each co}$$

Some outside inflow must close the gap — exactly the U.S. situation, where net international migration now accounts for most population growth.

The previous device: entrants compared value inside to an outside value $\bar{V}^{out}$ and entered with logit probability ($\lambda = 2$). Three problems:

1. λ is unidentified — and generated $\sim 90\%$ of the +9–10% policy population responses
2. The margin is nearly saturated ($q^* = 0.97$): responses are one-sided by construction
3. Holding the outside fixed while a *national* policy moves all inside values is a single-treated-city experiment in a one-market national model

The Quota Closure

Delete the choice layer; hold the flows themselves fixed:

$$S E_0 = \bar{R} S B_0(\text{policy}) + \bar{M}, \quad \bar{R} = \frac{(1 - s^{out}) E_0}{B_0} = 0.969, \quad \bar{M} = s^{out} E_0,$$

with $s^{out} = 0.169$ the outside-origin share of entrants. Institutions justify the invariance: U.S. immigration is quantity-rationed and oversubscribed — the marginal entrant is set by the quota, not by indifference.

- Baseline is **bit-identical** to the old one (the logit was reverse-engineered to this same accounting); zero recalibration
- Population responds through the one margin the model disciplines — births — with renewal multiplier $(1 - s^{out})/s^{out} = 4.92$, damped by housing scarcity
- Below-replacement fertility + a constant absolute inflow $\Rightarrow$ a stationary population exists and is stable (standard demography)

Policy Results

Funded Property-Tax and Grant Reforms

Raise the property tax from 1% to 2%; rebate lump-sum, or fund a purchase grant for family-sized homes. All changes vs. the funded 1% baseline, percent:

Arm	Policy	TFR	Births	Househ.	$-\Delta\text{Imm.}$	Price	Rent
Floor	2% tax	+0.38	+0.33	+1.70	1.74	-17.5	-6.1
Floor	tax + grant	+0.75	+0.67	+3.46	3.52	-17.3	-5.8
Tilt	2% tax	+0.20	+0.18	+0.90	0.92	-17.7	-6.3
Tilt	tax + grant	+0.49	+0.44	+2.23	2.26	-17.2	-5.7

In U.S. units: +0.33–0.67% births is roughly 12–24 thousand extra babies per year immediately; the household-count response (+1.7–3.5%) is the stationary limit, reached on a ~century timescale. The price falls because the tax lowers owner demand — the **down-payment barrier** drops 17.5% while renter rent (tax-inclusive) falls only 6%.

What the Population Numbers Mean

One number, three equivalent units — all renewal arithmetic on the birth response:

households $+3.5\%$ $\iff$ required immigration -3.5% at fixed population
 $\iff$ 3.5% of the fertility gap closed.

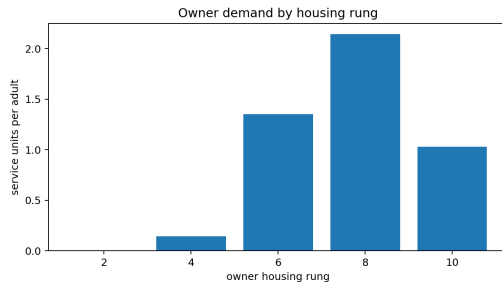
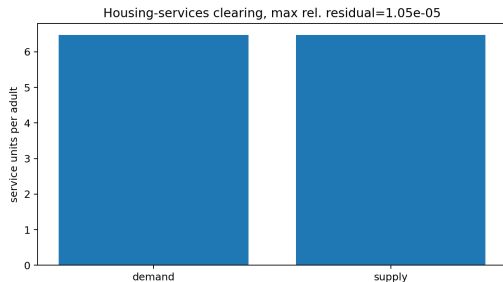
- **Housing feedback is real but small:** the arithmetic multiplier overstates the realized response by only 0.02–0.10pp — scarcity makes pronatal policy mildly self-limiting
- **Honest headline change:** $+0.9\text{--}3.5\%$, not the withdrawn $+9\text{--}10\%$ (that was λ)
- **Timing:** half the gap closes in $\sim$ a century; the immediate objects are the birth and required-immigration flows

Open Decisions

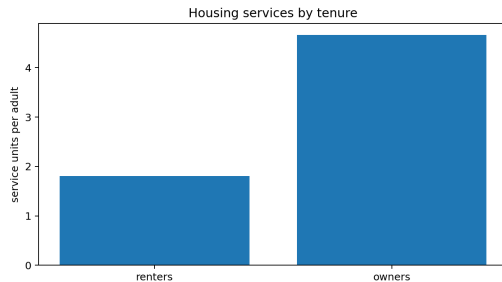
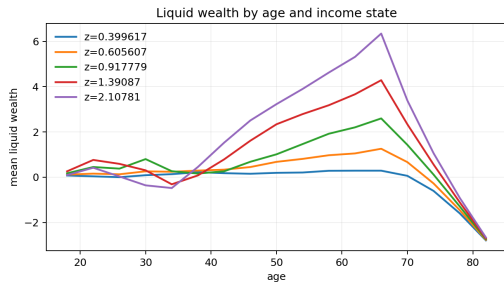
1. **The anchor concept.** New ACS measurement: recent immigrants are only 3.7% of age-18–22 household entrants — below the 14.3% feasibility floor. But immigrants mostly arrive *older* than the stylized entry age. Recommended: anchor $\bar{M}$ to *age-aggregated* net migration (~ 0.17 – 0.22 , brackets the provisional 0.169); the alternative is multi-age immigrant entry
2. **The fertility block.** Closure feasibility requires $B_0/E_0 \geq 0.831$; the repaired-E5 chain-6 winner fails it (screened out), and the misses that survive 24 chains — childlessness, old-age wealth dispersion, late first births — are mechanism questions, not calibration slack
3. **Scope.** Welfare comparisons stay per-household; the entry margin, if it ever needs to be behavioral, is household formation (headship), not an international logit
Population effects are now: births $\times$ transparent demography, damped by housing scarcity — conditional on fixed immigration.

Appendix

Appendix: Housing Market Diagnostics

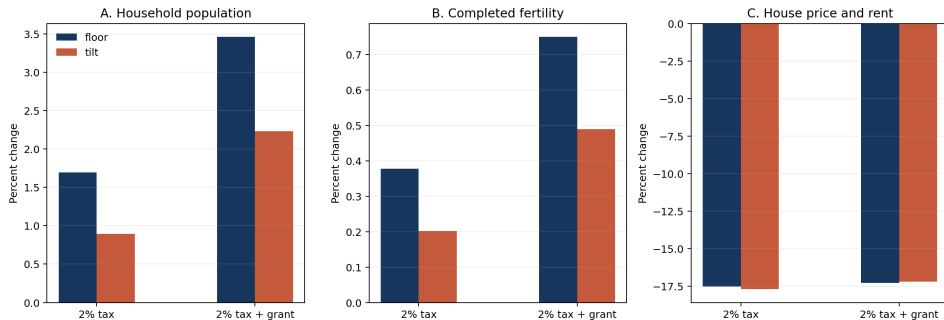


Appendix: Wealth and Tenure Diagnostics

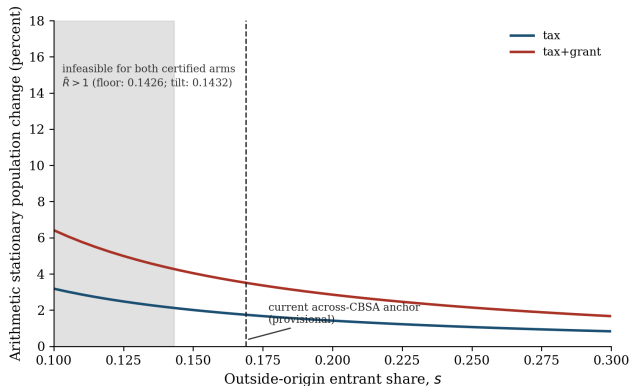


Appendix: Policy Comparison

Experimental repaired-E5 funded policies: quota population closure



Appendix: Anchor Sensitivity and Convergence Speed



Stationary population response as a function of the outside-origin share s^{out} , with the infeasible region ($s^{out} < 0.143$) shaded. Convergence: generational decay $\rho = \bar{R}B_0/E_0 = 0.831$ gives a half-life of 97–112 years ($g = 26$ –30); an age-structured variant with uniform birth timing gives ~ 230 years (upper bound).

Appendix: The Retired Logit as a Sensitivity Row

The old protocol's numbers survive as a labeled robustness row, not a model:

Arm	Policy	Quota population	Logit sensitivity ($\lambda = 2$, unidentified)
Floor	2% tax	+1.70	+9.98
Floor	tax + grant	+3.46	+10.10
Tilt	2% tax	+0.90	+8.99
Tilt	tax + grant	+2.23	+9.06

The gap between columns is what an unidentified taste parameter was contributing. The logit row bounds “what if entry responded to values” without pretending to identify it.

Appendix: National Entry-Anchor Measurement (ACS 2012–2023)

Measure (household heads unless noted)	Share	Multiplier $(1 - s)/s$
Ages 18–22, foreign-born, arrived $\leq 4y$ (headline)	3.71%	25.9
Ages 18–22, foreign-born, arrived $\leq 8y$	5.36%	17.7
Ages 18–25, foreign-born, arrived $\leq 4y$	3.47%	27.9
Ages 18–22, foreign-born stock (upper bound)	9.04%	10.1
Persons 18–22, foreign-born, arrived $\leq 4y$	2.89%	33.7
Quota feasibility floor	14.3%	—
Age-aggregated net-migration bracket (to verify)	17–22%	3.5–4.9

Every entry-age-window measure is infeasible; the age-aggregated concept is not. The two are different objects because immigrants mostly arrive above the stylized entry age — the anchor decision on the Open Decisions slide.

Appendix: Multi-Start Record (overnight)

16 fresh chains, unchanged 12-moment / 9-parameter contract, 225-minute budget each, strict-repeat certification:

	Best	Median	Worst
Fresh chains (seeds 09–24)	263.5	~284	370.5
Certified incumbent (Aug 5, chain 6)	249.186	— stands	

All 16 eligible with exact strict repeats on the run's best; market residuals $\leq 2.2 \times 10^{-5}$. With 24 chains total across the two arrays, the remaining misses are search-converged.